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Automated Side Mirror Car Wiper System

Department of Electronics and Telecommunication Engineering

Group Number : A16

Details of the Project Group

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Project Title: " Automated Side Mirror Car Wiper System".

Project Guide: Dr. M. S. Vanjale

Project Domain: Automation

Date	Week	Milestone for the week	Remark	Guide's Signature
05/07/2024	1	Group Formation and Project Idea Selection	Finalized group members and selected side mirror wiper automation as project.	
10/07/2024	2	Project Orientation lecture	Attended orientation on project planning and execution stages.	
22/07/2024	3	Initial Literature Survey	Reviewed existing automatic mirror and wiper systems.	

08/08/2024	4	Literature Review and Problem Definition	Defined design challenges in detecting rain and motor response.	
20/08/2024	5	System Design & Component Selection	Selected STM32, SG90 servo, YL-83 sensor, and basic power modules.	
30/08/2024	6	Hardware Design and Simulation	Simulated servo wiper logic and rain sensor response in software.	
12/09/2024	7	1st Project Presentation	Presented problem statement, solution approach, and initial design.	
20/09/2024	8	Component Functionality and Testing	Tested sensor output and motor PWM control.	
26/09/2024	9	Risk Analysis and Troubleshooting Plan	Planned measures for rain misdetection and false triggers.	
07/10/2024	10	Prototype Assembly	Mounted components on mirror structure; started integration.	
14/10/2024	11	2nd Project Presentation	Demonstrated sensor-to-servo response under demo rain.	
25/10/2024	12	Software Development and Integration	Finalized code structure for sensor reading and wiper activation.	
11/11/2024	13	Initial Testing and Debugging	Checked system reliability and wiper timing control.	
18/11/2024	14	Stage 1 Evaluation	Presented working prototype and system block diagram.	
02/12/2024	15	Code Optimization	Reduced delay and enhanced sensor threshold logic.	

09/12/2024	16	Voltage Regulator and Power Setup	Tested 7812 stability; verified power to sensor and board.	
16/12/2024	17	Testing	Test the wiper according to the rain intensity .	
23/12/2024	18	Data Logging Integration	Enabled real-time wiper activity	
30/12/2024	19	Wi-Fi Module Testing	Checked wireless sync between STM32 and mobile display app.	
06/01/2025	20	Testing in Simulated Rain Conditions	Tested setup using artificial rain; validated triggering accuracy.	
13/01/2025	21	Internal Review and System Tuning	Adjusted sensor delay and motor torque settings.	
20/01/2025	22	Bug Fixes and Code Refinement	Resolved servo oscillation during intermittent rain.	
27/01/2025	23	Report Format Finalization	Started documenting system structure and flowcharts.	
03/02/2025	24	Final Demo Preparation	Full system tested in final enclosure and vehicle mount.	
10/02/2025	25	Poster and PPT Design	Designed poster and presentation slides for Stage 2.	
17/02/2025	26	Guide Review and Approval	Submitted updated files to guide for approval.	
24/02/2025	27	Incorporating Feedback	Made suggested corrections in layout and testing notes.	
03/03/2025	28	Submission to Department	Submitted documents, code, and demonstration files.	
10/03/2025	29	Backup and Archiving	Archived final code and 3D models	

17/03/2025	30	Demo Setup Testing	Validated hardware wiring and performance before viva.	
31/03/2025	31	Final Adjustments	Improved wire management and added system labels.	
14/04/2025	32	PPT Review with Guide	Conducted mock presentation with mentor guidance.	
21/04/2025	33	Submission Verification	Verified all forms and report printouts.	
28/04/2025	34	Final Testing Under Load	Tested system for continuous operation and endurance.	
05/05/2025	35	Final Adjustments	Last-minute improvements in wiring, labeling. and user manual content.	
12/05/2025	36	Practice Presentation	Practiced delivery and Q&A round as preparation for final exam panel.	
02/06/2025	37	Final Testing Under Load	Tested system and varying input for durability.	
09/06/2025	38	Final Revision and Confidence Building	Reviewed complete system and documents.	